GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for Venturi Based Wet Gas Flow Meters- BBOU/ELEP**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-006
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Specification for Venturi Based Wet Gas Flow Meters- BBOU/ELEP

R02	19.10.22	Re-Issued for Review	P. Roy	P. Olukah	A. Duru	
R01	05.07.22	Issued for Review	P. Roy	P. Olukah	W. Irem	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	05.07.22	Issued for Review
R02	19.10.22	Re-issued for review after incorporating TCE comments on revision R01
A01		

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)

for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-006

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with ELEP & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS:

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

Abbreviations

GVF	Gas Volume Fraction
LVF	Liquid Volume Fraction
SCADA	Supervisory Control and Data Acquisition System
ISO	International Standard Organisation
CGR	Gas to Condensate Ratio
STB	Stick Tank Barrel
MMscf	Million standard cubic feet
BCPD	Barrels of Condensate per Day
BWPD	Barrels of Water per Day
SPDC	The Shell Petroleum Development Company of Nigeria Limited. They are the Feedstock Gas Supplier and responsible party for the Custody Transfer Metering Package

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (ELEP), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with ELEP & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

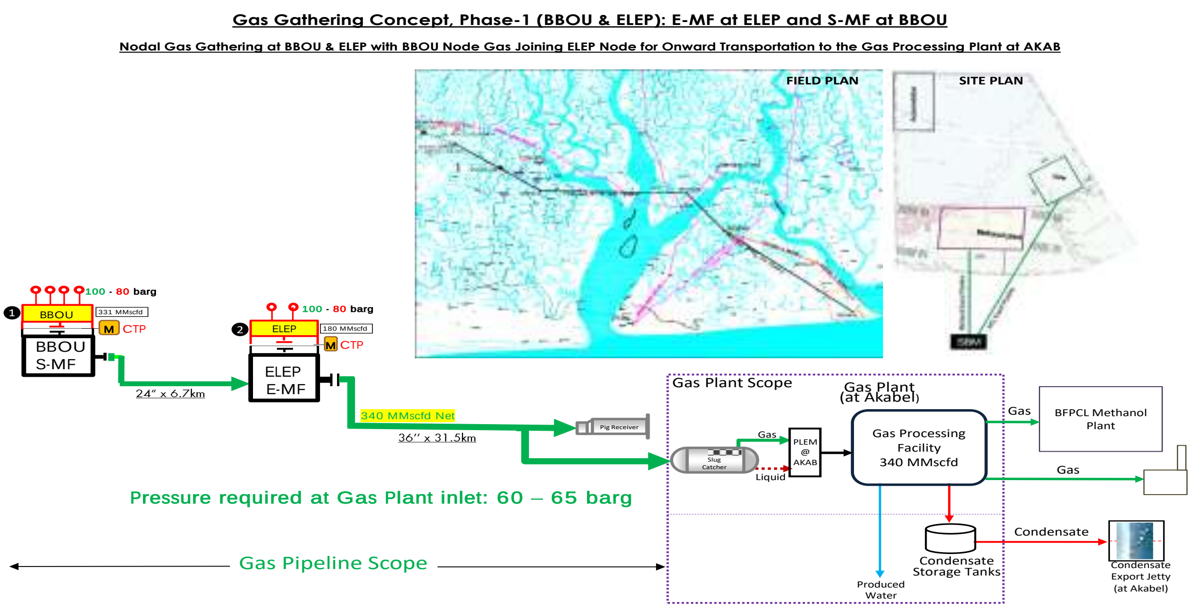

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification covers the minimum requirements for the design, manufacture, assembly, Inspection, testing, certification, delivery and software applications for Venturi Meter based wet gas metering for BBOU and ELEP.

2. General

2.1. Vendor shall study the requirement of wet gas meter carefully and suggest the best possible technology for the application.

2.2. Venturi based wet gas meters shall be designed based on ISO 5167, ISO 11583, ASME MFC 19G, ISO TR 12748 and other related international standards.

2.3. Field instrumentation shall have a minimum ingress protection of IP65 in accordance with IEC 60529. All instruments shall be intrinsically safe or explosion proof and shall be certified to specific hazardous area class by statutory body such as ATEX, PTB.

2.4. Materials of Construction: Materials of construction for instrument equipment shall be consistent with corrosion resistance and the pressure and temperature conditions of the process fluids involved. Generally, the parts in contact with the process fluid shall be of a material equal to or better than that used for the related vessel or line.

2.5. Pressure & temperature rating

Pressure and temperature rating of instruments and components shall conform to the design rating of the process system to which they are connected. Instruments which can be exposed to vacuum shall have under-range protection to full vacuum.

2.6. Hazardous area requirement: Classification of Hazardous Areas shall be Zone1, IIA/IIB, T4. All instrumentation located or partially located in the hazardous area shall be suitable for classified hazardous area. All instruments and Junction Boxes shall be certified as Explosion Proof (Exd). Power supply shall be switched off before opening the Instrument or Junction Box. ATEX/PTB approved certification shall be supplied for all electrical / electronic equipment in the hazardous area.

2.7. Accessibility: All instruments shall be accessible from grade or platform for measurement. In case of gas service where instruments are mounted above the taps, the installation of platform shall be considered, if accessibility is not available.

2.8. Vendor shall follow the international codes and standards and local regulations

2.9. Vendor shall offer inspection and testing as per the contractual requirement.

2.10. Vendor shall provide all related documentations like installation diagram, piping GA and instrument hook up diagram, loop diagram, data sheets for all instruments etc.

3. Technical Specification of Wet Gas Metering

3.1. The scope of venture based wet gas metering system for BBOU and ELEP shall consist of following

- Design of the wet gas metering system based on the process data provided in this specification.
- Supply of integrated assembly of venture meters along with Gauge Pressure Transmitters (PT), Differential Pressure Transmitters (DPT), Temperature Transmitter with RTD and TW, First isolation valve for pressure and differential transmitters, Double Block and Bleed Valve Pressure and differential pressure transmitters, impulse piping and fitting complete with installation of all transmitters.
- Low powered Field mounted Flow Computer (Hazardous Area) along with all Input/Output modules that shall interfaces with the PT, DPT and TT from the venturi measurement. All necessary software shall be supplied and configured.
- Interface with SCADA system through serial links.

3.2. All instruments shall be installed within the vendor supplied venture meter or extended spool, if required without any dependency of the user.

3.3. Entire system shall be calibrated at a reputed laboratory like CEESI, SWRI.NEL etc. Calibration shall be done at 5 points and calibration report shall be provided. Customer shall have rights to witness the calibration.

3.4. Vendor shall suggest all utilities, power and pre-requisites that are required from the purchasers

3.5. Cabling between all transmitters and computers shall be done by the vendor. Cabling between transmitters/ flow computer to host SCADA shall be by purchaser.

3.6. Vendor shall suggest upstream and downstream requirement of Venturi based wet gas meter.

Tagwise datasheets including process parameters shall be provided during Detailed Engineering Design when Vendor inputs would have been obtained.

4. Functional Specification for Wet Gas Metering

4.1. The pressure loss across a differential pressure (DP) device is a highly complex function of the properties of the fluid stream for multiphase flow. By measuring the total differential pressure (DPt) across the Venturi and the standard Venturi differential pressure (DPv), the pressure loss ratio (PLR= DPt/DRv) can be calculated. Vendor shall use the PLR together with gas to condensate ratio (CGR) to measure multi-phase flow of rates of gas, condensate and water in real time.

4.2. For BBOU, one (1) no. Pressure Transmitter, two (2) no.s of Differential Pressure Transmitter and a temperature transmitter shall be used. BBOU shall measure high Gas Volume Fraction (GVF).

4.3. For ELEP, one (1) no. Pressure Transmitter, three (3) nos. of Differential Pressure Transmitter and a temperature transmitter shall be used. ELEP shall measure high Liquid Volume Fraction (LVF).

4.4. **Typical Operating Range**: Turndown >8:1, Gas mass fraction 50-100% Water liquid ratio 0-100%

4.5. **Typical Performance**: Typical Uncertainty (95% Confidence Level): • Gas mass flow rate ±2% • Condensate mass flow rate ±10% • Water volume flow rate ±1 am^3/h

4.6. **Typical Repeatability**: Gas mass flow rate <0.40%, Condensate mass flow rate <2.0%, Water sensitivity ±0.2 am^3/h

4.7. **Permanent Pressure Loss**: Pressure loss specific to application (< 1 bar)

5. Venturi Meter

5.1. Venturi meter shall be used for the wet gas metering. Material of Construction of venture meter shall be SS316.

5.2. Venturi meter shall be designed as per ISO-5167, ISO 11583, ASME MFC 19G, ISO TR 12748 and other related international standards.

5.3. Sizing of the meter shall be responsibility of the vendor and provide the piping interface details.

6. Temperature Transmitter

6.1. Temperature Elements

6.1.1. Resistance Thermometer Detectors (RTD's) Resistance thermometers with duplex 3-wire type, Platinum 100 ohm at 0 °C complying with IEC-60751 shall be used.

6.2. Transmitters

6.2.1. Transmitters should be 4-20 MA,Smart HART type.

6.2.2. Transmitter shall be integrally mounted with RTD.

6.2.3. Accuracy of the Temperature Transmitter shall be +/-0.08 degC.

6.2.4. In the event of temperature element failure, transmitters shall be arranged to drive towards the alarm condition or safe control condition where one exists, or otherwise upscale

6.3. Protecting Thermowells

6.3.1. Thermowell shall be machined from bar stock material. Other materials shall be furnished as required in the piping specification. Temperature elements shall be supplied complete with thermowell, extension, head and integrally mounted transmitter as a complete assembly. All thermowells shall be SS316L and meet the Karman Wake Frequency calculation as per PTC 19.3. If it fails to achieve the wake frequency calculation, collar or vortex shedding design shall be used to meet the velocity criteria. Thermowell ts design and geometry shall meet the requirements of Part II, Section 3.9 of DEP 31.38.01.24-Gen. At all process conditions, the ratio of the Strauhal frequency and the thermowell natural frequency shall be less than 0.8, as stated in ASME PTC 19.3.

6.3.2. Thermowells shall be flanged according to equipment and pipe specifications. The flange size shall be 40 mm (1.1/2 inch) with rating and facing according to the line classification.

6.3.3. According to API RP 551, the immersion length of thermowells in the pipes (the length from well tip to pipe wall) shall be maximum 5" or 125mm. The immersion length in vessels liquid shall be 400 mm. In vessels containing gas, the immersion length shall be according to the table below:

Vessel diameter (D)	Thermowell immersion length(mm)
D upto 800 mm	200
800<D≤1200 mm	300

The insertion length depends on the length of the nozzle or pipe fitting, taking any possible refractory lining into account.

If the thermowell design fails to pass the wake frequency with the give gas velocity even after increasing the tip diameter or other mechanical parameters, vortex shedding type helical thermowell shall be used. Velocity collar shall be avoided.

7. Pressure and Diffrential Pressure Instruments

7.1. Pressure Transmitters

7.1.1. Pressure transmitters shall be referenced to atmospheric pressure.

7.1.2. Transmitters should be 4-20 mA, smart HART type with LCD display, with independent adjustment of zero and span at the instrument and from a remote location. Accuracy shall be +/-0.075% or better of the calibrated span. Vendor shall decide the accuracy to meet the overall performance of the wet has metering.

7.1.3. Materials of construction shall be suitable for the process media and based upon the piping specification. In general pressure transmitters should have Aluminium Enclosure with low copper and stainless-steel wetted pats.

7.1.4. Scales shall be selected to cover the range requirements for both start-up, normal and design conditions of plant operation.

8. Flow Computer

8.1. Field mounted, hazardous area certified flow computer shall be supplied along with the venturi and other instruments.

8.2. Flow computer shall meet ISO 11583 and shall be approved from relevant authority.

8.3. Flow computer shall accept all signal inputs like pressure transmitter, differential pressure transmitter, temperature transmitter.

9. Tag Nos.

9.1. Tag Nos. for ELEP wet gas metering shall be

- **Venturi Meter** – ELEP-FE-1001M
- **Gauge Pressure Transmitter**- ELEP-PIT-1001M
- **Differential Pressure Transmitter**- ELEP-DPIT-1001M and ELEP-DPIT-1002M
- **Temperature Element and Temperature Transmitter**- ELEP-TE-1001M and ELEP-TIT-1001M
- **Flow Computer**- ELEP-FC-1001M

9.2. Tag Nos. for ELEP wet gas metering shall be

- **Venturi Meter** – ELEP-FE-1001M
- **Gauge Pressure Transmitter**- ELEP-PIT-1001M
- **Differential Pressure Transmitter**- ELEP-DPIT-1001M,ELEP-DPIT-1002M, ELEP-DPIT-1003M
- **Temperature Element and Temperature Transmitter**- ELEP-TE-1001M and ELEP-TIT-1001M
- **Flow Computer**- ELEP-FC-1001M

9.3. Pressure and Temperature Conditions

- **Pressure: 80-100 barg**
- **Temperature: 50-95 degC**

10. Production Forecast & Fluid Characteristics

10.1.1. BUBOUWE BOU

Condensate-Gas Ratio (CGR) is 5.56 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWPD)
1	181	944	907
2	181	878	907
3	181	788	907
4	181	742	7855
5	181	707	7547
6	215	840	8491
7	305	1217	**9002**
8	327	1323	8193
9	**331**	**1338**	8038
10	67	239	2397
11	21	59	1031
12	16	42	784
13	2	4	85
14	0	0	0
15	0	0	0
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0
TCQ	800	3	21
	(BSCF)	(MMBBL)	(MMBBL)

10.1.2. ELEPA

Condensate-Gas Ratio (CGR) is 60.64 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWPD)
1	**180**	**10088**	900
2	180	10088	900
3	180	10088	900
4	180	10088	7980
5	180	10088	**7666**
6	145	8146	4902
7	52	2911	2597
8	29	1630	1454
9	25	1386	1236
10	19	1076	960
11	11	628	561
12	9	478	426
13	1	52	46
14	0	0	0
15	0	0	0
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0
TCQ	435	24	11
	(BSCF)	(MMBBL)	(MMBBL)

11. Gas Composition

11.1.1. Gas Compositions from Nominated Fields

11.1.1.1. Field Compositions

Field1: Bubouwe Bou Analogue		Field 2: Elepa		Field 3: Nun River H4.0 (Future)	
Component	Mol.%	Component	Mol.%	Component	Mol.%
N_2	0.07	N_2	0	N_2	NA
CO_2	0.59	CO_2	2.22	CO_2	0.77
C_1	92.92	C_1	81.66	C_1	72.83
C_2	3.16	C_2	6.13	C_2	7.68
C_3	1.01	C_3	2.59	C_3	4.85
iC_4	0.61	iC_4	0.75	iC_4	1.46
nC_4	0.44	nC_4	0.81	nC_4	1.97
iC_5	0.25	iC_5	0.45	iC_5	0.95
nC_5	0.19	nC_5	0.33	nC_5	0.76
C_6	0.23	C_6	0.65	C_6	1.03
C_7	0.19	C_7	0.9	C_7	1.41
C_8	0.04	C_8	0.87	C_8	1.61
C_9	0.03	C_9	0.38	C_9	0.9
C_{10}	0.02	C_{10}	0.3	C_{10}	0.65
C_{11}	0.03	C_{11}	0.23	C_{11}	0.47
C_{12+}	0.22	C_{12}	0.22	C_{12}	0.37
C_{13}	NA	C_{13}	0.23	C_{13}	0.34
C_{14}	NA	C_{14}	0.22	C_{14}	0.3
C_{15}	NA	C_{15}	0.19	C_{15}	0.26
C_{16}	NA	C_{16}	0.15	C_{16}	0.21
C_{17}	NA	C_{17}	0.12	C_{17}	0.16
C_{18}	NA	C_{18}	0.11	C_{18}	0.14
C_{19}	NA	C_{19}	0.06	C_{19}	0.08
C_{20+}	NA	C_{20+}	0.43	C_{20+}	0.8
TOTAL	100	TOTAL	100	TOTAL	100
Flash CGR	7	Flash CGR	59.7	Flash CGR	161.7

11.1.1.2.Portfolio Composition

Range for Portfolio	Lean Sample		Rich Sample
Component	Mol.%		Mol.%
N_2	0.07		NA
CO_2	0.59		0.77
C_1	92.92		72.83
C_2	3.16		7.68
C_3	1.01		4.85
iC_4	0.61		1.46
nC_4	0.44		1.97
iC_5	0.25		0.95
nC_5	0.19		0.76
C_6	0.23		1.03
C_7	0.19		1.41
C_8	0.04		1.61
C_9	0.03		0.9
C_{10}	0.02		0.65
C_{11}	0.03		0.47
C_{12+}	0.22		0.37
C_{13}	NA		0.34
C_{14}	NA		0.3
C_{15}	NA		0.26
C_{16}	NA		0.21
C_{17}	NA		0.16
C_{18}	NA		0.14
C_{19}	NA		0.08
C_{20+}	NA		0.8
TOTAL	100		100

11.1.1.3. Well Compositions

11.1.1.3.1. BBOU D3

As used in simulation model - Tuned PVTsim Model using Bonn-31ST PVT as analogue.

Component	Mol %	Molecular Weight	Liquid Density (g/cc)
N2	0.070	28.014	
CO2	0.590	44.010	
C1	92.922	16.043	
C2	3.160	30.070	
C3	1.010	44.097	
iC4	0.610	58.124	
nC4	0.440	58.124	
iC5	0.250	72.151	
nC5	0.190	72.151	
C6	0.230	86.178	0.664
C7	0.190	96.000	0.738
C8	0.040	107.000	0.765
C9	0.030	121.000	0.781
C10-C11	0.050	141.800	0.794
C12	0.068	161.000	0.800
C13	0.047	175.000	0.805
C14	0.032	190.000	0.808
C15	0.022	206.000	0.812
C16	0.015	222.000	0.815
C17	0.010	237.000	0.818
C18	0.007	251.000	0.821
C19-C44	0.016	293.067	0.830
Total	100.000		

Bonny 31ST Reservoir Fluid Composition				
No	Component	Flash Gas	Flash Oil	Reservoir Fluid
		[mol %]	[mol %]	[mol %]
1	N_2	0.07	0	0.07
2	CO_2	0.59	0	0.59
3	C_1	93.26	0	92.92

4	C_2	3.17	0.01	3.16
5	C_3	1.01	0.03	1.01
6	i-C_4	0.61	0.07	0.61
7	n-C_4	0.44	0.07	0.44
8	i-C_5	0.25	0.16	0.25
9	n-C_5	0.19	0.16	0.19
10	C_6	0.22	1.88	0.23
11	C_7	0.17	6.21	0.19
12	C_8	0.02	6.61	0.04
13	C_9	0	7.13	0.03
14	C_{10}	0	5.7	0.02
15	C_{11}	0	7.83	0.03
16	C_{12+}	0	64.14	0.22

	Total		**100**	**100**	**100**
	Density @ 60F	[g/cm^3]		0.811	
	MW	[g/mol]	17.97	165.61	18.47
	GOR	[Scf/Stb]			180,011.90
	Gravity	[air=1]	0.62		
	C_{7+}	Mole %	0.45	97.62	0.53
	C_{7+} MW	[g/mol]	100.89	167.69	143.17
	C_{12+} MW	[g/mol]	-	191.6	191.6
	CGR (bbls/mmscf)				5.56

11.1.1.3.2.Elepa U-4

Component	Res Fluid	Liquid		Gas	
	(z)		(x)		(y)
Totals	mol%		mol%		mol%
N2	0.0000		0.00		0.00
CO2	2.2200		0.00		2.30
H2S	0.0000		0.00		0.00
c1	81.6600		0.00		84.63
c2	6.1300		0.03		6.35
c3	2.5900		0.27		2.67
ic4	0.7500		0.37		0.76

Component	Res Fluid	Liquid		Gas	
nc4	0.8100		0.54		0.82
ic5	0.4500		0.87		0.43
nc5	0.3300		0.68		0.32
c6	0.6500		2.26		0.59
c7	0.9000		6.75		0.69
c8	0.8700		12.76		0.44
c9	0.3800		10.87		0.00
c10	0.3000		8.62		0.00
c11	0.2300		6.64		0.00
c12	0.2200		6.15		0.00
c13	0.2300		6.45		0.00
c14	0.2200		6.16		0.00
c15	0.1900		5.51		0.00
c16	0.1500		4.37		0.00
c17	0.1200		3.36		0.00
c18	0.1100		3.02		0.00
c19	0.0600		1.62		0.00
c20+	0.4300		12.70		0.00
Totals	100.00		100.00		100.00

11.1.1.3.3.Elepa U-7

Component	Res Fluid	Liquid		Gas	
	(z)		(x)		(y)
Totals	mol%		mol%		mol%
co2	2.1400		0.00		2.16
N	0.0000		0.00		0.00
c1	86.6700		0.00		87.30
c2	6.2100		0.37		6.25
c3	2.0500		0.24		2.06
ic4	0.5200		0.16		0.52
nc4	0.4900		0.26		0.49
ic5	0.2500		0.39		0.41
nc5	0.1600		0.36		
c6	0.2800		1.88		0.27

Component	Res Fluid	Liquid		Gas	
c7	0.3700		5.97		0.33
c8	0.3100		13.52		0.21
c9	0.1000		13.33		0.00
c10	0.0700		10.03		0.00
c11	0.0500		7.46		0.00
c12	0.0500		6.75		0.00
c13	0.0500		6.70		0.00
c14	0.0400		6.01		0.00
c15	0.0400		4.99		0.00
c16	0.0300		4.02		0.00
c17	0.0200		3.00		0.00
c18	0.0200		2.84		0.00
c19	0.0200		2.23		0.00
c20+	0.0700		9.49		0.00
Totals	100.01		100.00		100.00

12. Recommended Vendors

12.1. Vendor shall be selected from CLIENT's approved Vendors List.

12.2. Without prejudice to the final CLIENT vendor list at the time of instrument purchase, the following additional vendors are recommended for consideration.

Solartron **Dual Stream 1** and Solartron **Dual Stream 2**, or as may be recommended by SPDC for their custody transfer metering package.

Krohne

Tektrol

Or Equivalent

Any one of above vendors not on CLIENT's approved vendor list shall go through the CLIENT's pre-qualification process before being considered.